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# MIT SCHOOL OF COMPUTING

**Class : LY**
**Group Id: BCAIAA61**

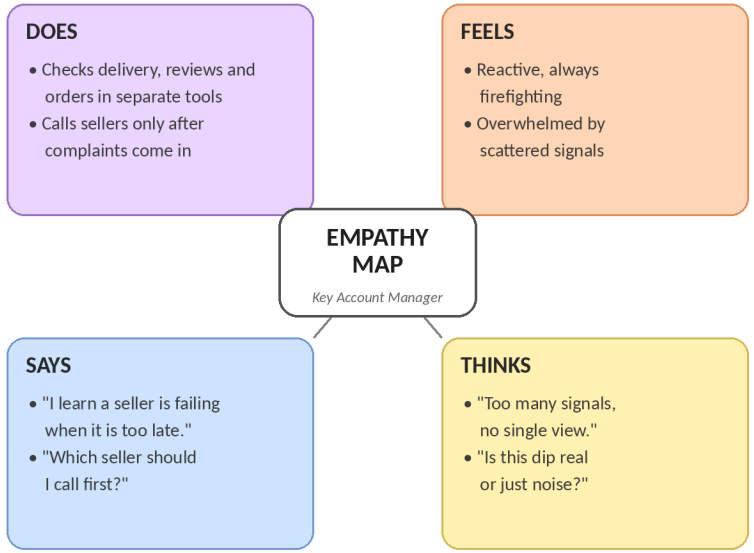
## DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

# *Seller Health & Intervention Platform*

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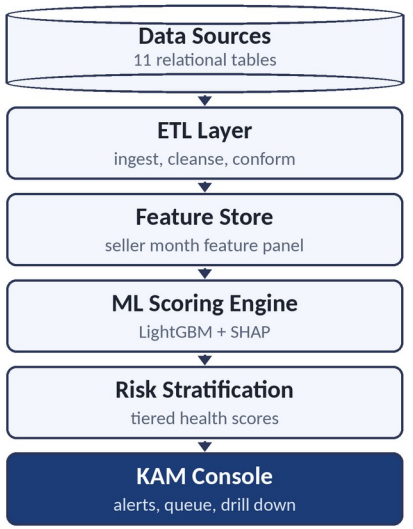
**Faculty Guide:**

**Prof. Sneha Singha**




| SR. NO. | Requirement | Proposed Solution |
|---|---|---|
| 1. | Detect seller deterioration early | Risk model validated on future 90 day outcomes |
| 2. | Make every score auditable | SHAP attribution with pillar level breakdown |
| 3. | Isolate seller vs carrier delay | Timestamp split of dispatch SLA vs carrier transit |
| 4. | Prioritize account manager effort | Intervention queue ranked by risk × GMV at stake |

## Block Diagram




## Problem statement

Seller performance data is fragmented across delivery, review, order, and payment systems. Issues surface only after complaints or revenue loss, with no unified, explainable signal of which sellers are at risk and why.

## Proposed Solution

A machine learning based Seller Health Index built on 100k+ marketplace orders. An ETL pipeline integrates 11 tables into a seller month feature panel; a gradient boosted model on rolling 90 day windows yields calibrated risk scores with SHAP driver attribution, served via a dashboard with tier alerts, a ranked intervention queue, and order level drill down.

## Scope and Feasibility

1.Uses the public Olist ecommerce and marketing funnel datasets; fully reproducible.
2.Scores validated by walk forward backtesting against realized 90 day outcomes.
3.Intervention estimates are correlational; causal inference is future work.